TESTING REPORT

# PlaceIQ

Verification of the database, business rules, queries and web API: 91 of 91 automated tests passing

**Fundamentals of Database Systems (FDS)**

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# Contents

1. Introduction ..... 3
1.1 Objectives ..... 3
2. Test strategy ..... 4
2.1 Test levels ..... 4
2.2 Approach ..... 4
2.3 Expected error codes ..... 4
3. Test environment ..... 6
4. Results summary ..... 7
4.1 First run ..... 7
5. Requirements traceability ..... 8
6. Detailed test cases ..... 10
6.1 Schema structure ..... 10
6.2 Declarative constraints ..... 10
6.3 Triggers and procedures ..... 11
6.4 Views and queries ..... 11
6.5 Data invariants ..... 12
6.6 Transactions & concurrency ..... 12
6.7 Performance ..... 12
6.8 Security ..... 13
6.9 REST API ..... 13
7. Focus areas ..... 14
7.1 Concurrency (TC-TXN-02, TC-TXN-03) ..... 14
7.2 Performance (TC-PRF-01 to 03) ..... 14
7.3 Security (TC-SEC-01 to 09) ..... 14
7.4 Query correctness ..... 14
8. Defect and issue log ..... 15
9. Running the tests ..... 16
10. Conclusion and limitations ..... 17
Not covered ..... 17

*(If page numbers are missing, right-click the table and choose Update Field.)*

# 1. Introduction

This report describes how PlaceIQ was verified: what was tested, how, in which environment, and with what result. All results were produced by the automated test suite in tests/, executed with pytest against a real PostgreSQL 18 database rebuilt from the project scripts. Every table of results in this document is generated from the suite's own output file (tests/results/results.json), not typed by hand.

## 1.1 Objectives

- Show that every business rule (BR-1 to BR-15) is enforced by the database, including when the application is bypassed.
- Show that the analytical queries and views return correct answers, using independent cross-checks.
- Show ACID behaviour, including correct isolation under real concurrency.
- Measure, rather than assume, the effect of indexes.
- Show that SQL injection is prevented and that the demonstration path is safely sandboxed.
- Record honestly every defect found during development and testing.

# 2. Test strategy

## 2.1 Test levels

| Level | What is checked | Tests |
|---|---|---|
| Schema structure | The catalog contains exactly the designed objects: tables, keys, indexes, views, triggers, routines. | 10 |
| Declarative constraints | Each business rule enforced by the DDL rejects invalid data, and boundary values are accepted. | 20 |
| Triggers and procedures | Multi-table rules: eligibility, offer rules, the one-offer cascade, deadlines, expiry. | 16 |
| Views and queries | The five analytical queries and both views return provably correct answers. | 9 |
| Data invariants | The whole seeded dataset satisfies every rule and is reproducible. | 9 |
| Transactions & concurrency | Atomicity, isolation under two real concurrent sessions, deferred consistency at COMMIT. | 4 |
| Performance | Index effectiveness measured with EXPLAIN ANALYZE on 600,000 rows. | 3 |
| Security | Least-privilege role (DCL) and SQL-injection behaviour, sandboxing included. | 9 |
| REST API | Endpoints return correct data and surface database rule violations clearly. | 11 |

## 2.2 Approach

- **Known starting state.** The session begins by rebuilding the database from 02_ddl.sql, 03_logic.sql and 04_data.sql. The seed is deterministic (setseed), so expected values such as '8 students match job 25' are stable.
- **Isolation between tests.** Each test receives a connection whose outer transaction is rolled back afterwards; nothing a test does persists.
- **Many violations per test.** Expected failures run inside a SAVEPOINT (helper expect_error) and assert the exact SQLSTATE and message, so one test can check several violations and continue.
- **Committing tests.** Tests that must commit (two-session concurrency, API writes) use a fixture that rebuilds the seed afterwards.
- **Independent oracles.** Query results are verified by a different formulation (for example relational division by double NOT EXISTS versus by counting) or by recomputing the expected value directly.
- **Negative and positive cases.** Each constraint test also checks that a legal boundary value is accepted (for example CGPA exactly 10, back-to-back interview slots).

## 2.3 Expected error codes

| SQLSTATE | Meaning | Raised by |
|---|---|---|
| 23514 | check_violation | CHECK constraints; trigger RAISE … USING ERRCODE = 'check_violation' |
| 23503 | foreign_key_violation | Foreign keys (including the composite ones) |
| 23505 | unique_violation | UNIQUE / alternate keys |
| 23P01 | exclusion_violation | The interview double-booking EXCLUDE constraint |
| 23001 | restrict_violation | ON DELETE RESTRICT |
| 23000 | integrity_constraint_violation | Deferred totality trigger |

| SQLSTATE | Meaning | Raised by |
|---|---|---|
| 25006 | read_only_sql_transaction | Write attempted inside READ ONLY (injection lab) |
| 42501 | insufficient_privilege | placeiq_readonly role |
| 55P03 | lock_not_available | lock_timeout while waiting for the student row lock |

# 3. Test environment

| Component | Version / detail |
|---|---|
| Database | PostgreSQL 18.6 on x86_64-windows |
| Python | 3.12.7 |
| pytest | 8.3.3 |
| psycopg | 3.2.9 |
| Operating system | Windows 11 |
| Database name / schema | placeiq / placeiq |
| Run started (UTC) | 2026-10-28 18:25:22 |
| Command | app\.venv\Scripts\python -m pytest tests -v |

# 4. Results summary

**Result**

91 of 91 tests passed, 0 failed, 0 skipped. Total test time 10.0 s (excluding database rebuilds). The suite was run twice consecutively with identical results.

**Table 1.** *Results by module*

| Module | Tests | Passed | Failed | Time (ms) |
|---|---|---|---|---|
| Schema structure | 10 | 10 | 0 | 102 |
| Declarative constraints | 20 | 20 | 0 | 247 |
| Triggers and procedures | 16 | 16 | 0 | 326 |
| Views and queries | 9 | 9 | 0 | 271 |
| Data invariants | 9 | 9 | 0 | 1,199 |
| Transactions & concurrency | 4 | 4 | 0 | 1,819 |
| Performance | 3 | 3 | 0 | 3,019 |
| Security | 9 | 9 | 0 | 340 |
| REST API | 11 | 11 | 0 | 2,677 |

## 4.1 First run

The first execution of the suite reported 87 passed and 4 failed. All four were analysed before anything was changed: three were defects in the tests themselves (D-08, D-09) and one revealed that a claim in the demo script was inaccurate on PostgreSQL 18 (D-07). None was a defect in the schema or the business logic. After the corrections the suite passed 91 of 91, and passed again on an immediate re-run.

# 5. Requirements traceability

Every business rule is covered by at least one test; most by several (a negative case at the constraint level and a data-wide invariant).

**Table 2.** *Requirement → test cases*

| Requirement | Test cases | Count |
|---|---|---|
| BR-2 | TC-CON-03 | 1 |
| BR-3 | TC-CON-01 | 1 |
| BR-4 | TC-CON-04, TC-CON-05, TC-DAT-08 | 3 |
| BR-5 | TC-CON-06 | 1 |
| BR-6 | TC-CON-07 | 1 |
| BR-7 | TC-CON-08, TC-CON-09 | 2 |
| BR-8 | TC-CON-10, TC-DAT-04, TC-SCH-06 | 3 |
| BR-9 | TC-CON-12 | 1 |
| BR-10 | TC-CON-13 | 1 |
| BR-11 | TC-DAT-05, TC-TRG-01, TC-TRG-02, TC-TRG-03, TC-TRG-04, TC-TRG-05, TC-TRG-06, TC-TRG-07 | 8 |
| BR-12 | TC-DAT-03, TC-TRG-10, TC-TRG-12, TC-TRG-13, TC-TRG-14 | 5 |
| BR-13 | TC-CON-14 | 1 |
| BR-14 | TC-QRY-02, TC-TRG-11 | 2 |
| BR-15 | TC-CON-15 | 1 |
| API | TC-API-01, TC-API-02, TC-API-03, TC-API-04, TC-API-05, TC-API-06, TC-API-07, TC-API-08, TC-API-09, TC-API-11 | 10 |
| Atomicity | TC-TXN-01 | 1 |
| Business logic | TC-SCH-09, TC-SCH-10 | 2 |
| Consistency | TC-TXN-04 | 1 |
| Data | TC-DAT-01, TC-DAT-02, TC-DAT-06, TC-DAT-09 | 4 |
| Data load | TC-SCH-07 | 1 |
| DCL | TC-SEC-01, TC-SEC-02, TC-SEC-03, TC-SEC-04 | 4 |
| Defence in depth | TC-SEC-08 | 1 |
| Design | TC-SCH-01 | 1 |
| Domain | TC-CON-02, TC-CON-11, TC-CON-19, TC-CON-20 | 4 |
| Domain integrity | TC-SCH-05 | 1 |
| Entity integrity | TC-SCH-02 | 1 |
| Function | TC-TRG-16 | 1 |
| Indexing | TC-PRF-01, TC-PRF-02, TC-PRF-03 | 3 |
| Injection | TC-SEC-05, TC-SEC-06, TC-SEC-07 | 3 |
| Isolation | TC-TXN-02, TC-TXN-03 | 2 |
| Offer rule | TC-DAT-07, TC-TRG-08, TC-TRG-09 | 3 |
| Parameterisation | TC-SEC-09 | 1 |
| Physical design | TC-SCH-04 | 1 |

| Requirement | Test cases | Count |
|---|---|---|
| Procedure | TC-TRG-15 | 1 |
| Q1 | TC-QRY-04 | 1 |
| Q2 | TC-QRY-05, TC-QRY-06 | 2 |
| Q3 | TC-QRY-07 | 1 |
| Q4 | TC-QRY-08 | 1 |
| Q5 | TC-QRY-09 | 1 |
| Queries | TC-QRY-01 | 1 |
| Referential actions | TC-CON-16, TC-CON-17, TC-CON-18 | 3 |
| Referential integrity | TC-SCH-03 | 1 |
| UI | TC-API-10 | 1 |
| Views | TC-QRY-03, TC-SCH-08 | 2 |

# 6. Detailed test cases

## 6.1 Schema structure

The catalog contains exactly the designed objects: tables, keys, indexes, views, triggers, routines.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-SCH-01 | Design | The schema contains exactly the 22 designed tables | PASS | 10.1 |
| TC-SCH-02 | Entity integrity | Every table has a primary key | PASS | 8.4 |
| TC-SCH-03 | Referential integrity | All 30 designed foreign keys exist<br>Includes the composite FKs used for disjointness, the sponsor category, the recruiter-company consistency rule and the interview aggregation. | PASS | 6 |
| TC-SCH-04 | Physical design | Every foreign key is backed by an index that can find its rows<br>PostgreSQL does not index FK columns automatically; an unindexed FK makes joins and ON DELETE checks scan the whole child table. Covered means: an index whose leading columns are exactly the FK columns, or a UNIQUE index on a leading subset of them (e.g. PK job_id covers FK (job_id, role_type)). | PASS | 7.6 |
| TC-SCH-05 | Domain integrity | At least 40 CHECK constraints guard attribute domains | PASS | 5.7 |
| TC-SCH-06 | BR-8 | The interview double-booking EXCLUDE constraint exists | PASS | 4.5 |
| TC-SCH-07 | Data load | Identity sequences were advanced past the explicitly inserted ids | PASS | 36.4 |
| TC-SCH-08 | Views | Both views (v_job_details, v_placement_status) exist | PASS | 9.4 |
| TC-SCH-09 | Business logic | All 9 business-rule triggers are installed | PASS | 7.8 |
| TC-SCH-10 | Business logic | The stored function and both procedures exist | PASS | 6 |

## 6.2 Declarative constraints

Each business rule enforced by the DDL rejects invalid data, and boundary values are accepted.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-CON-01 | BR-3 | CGPA outside 0-10 is rejected by a CHECK constraint | PASS | 14.1 |
| TC-CON-02 | Domain | Malformed or upper-case e-mail addresses are rejected | PASS | 4.6 |
| TC-CON-03 | BR-2 | A duplicate roll number is rejected (UNIQUE alternate key) | PASS | 13.9 |
| TC-CON-04 | BR-4 | A full-time job cannot also be given an internship row (disjoint)<br>The composite FK (job_id, role_type) only matches when role_type = 'INTERNSHIP'. | PASS | 9.8 |
| TC-CON-05 | BR-4 | A job without any subtype row fails the deferred totality check<br>Inserting the job and its subtype in the same transaction passes. | PASS | 17.1 |
| TC-CON-06 | BR-5 | A recruiter cannot post a job for a company they do not work for | PASS | 8.7 |
| TC-CON-07 | BR-6 | A student cannot apply to the same job twice | PASS | 23.9 |
| TC-CON-08 | BR-7 | An interview in a round the job does not have is rejected (FK to weak entity) | PASS | 17.9 |
| TC-CON-09 | BR-7 | A student cannot be interviewed for a job they never applied to (aggregation FK) | PASS | 10.8 |
| TC-CON-10 | BR-8 | Overlapping interview slots for one student are rejected; back-to-back slots are allowed | PASS | 13.2 |
| TC-CON-11 | Domain | An interview that ends before it starts is rejected | PASS | 5.3 |
| TC-CON-12 | BR-9 | A second offer on the same application is rejected | PASS | 9.8 |

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-CON-13 | BR-10 | ACCEPTED or DECLINED offers must carry responded_at | PASS | 8.3 |
| TC-CON-14 | BR-13 | A company cannot claim a sponsor row typed ALUMNUS (category / union type) | PASS | 6 |
| TC-CON-15 | BR-15 | Deleting a student who holds an offer is blocked (CASCADE then RESTRICT) | PASS | 28.8 |
| TC-CON-16 | Referential actions | Deleting a student with no offers cascades to all dependent rows | PASS | 23 |
| TC-CON-17 | Referential actions | Deleting a recruiter nulls job_role.posted_by but keeps the job | PASS | 16.3 |
| TC-CON-18 | Referential actions | A program that still has students cannot be deleted | PASS | 7.1 |
| TC-CON-19 | Domain | Fixed pay above total CTC is rejected | PASS | 3.9 |
| TC-CON-20 | Domain | A placement cycle that ends before it starts is rejected | PASS | 4.5 |

## 6.3 Triggers and procedures

Multi-table rules: eligibility, offer rules, the one-offer cascade, deadlines, expiry.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-TRG-01 | BR-11 | Applying with a CGPA below the job's cut-off is rejected by the trigger | PASS | 10.7 |
| TC-TRG-02 | BR-11 | A student who opted out of placements cannot apply | PASS | 9.3 |
| TC-TRG-03 | BR-11 | Applying after the application deadline is rejected | PASS | 12.2 |
| TC-TRG-04 | BR-11 | A student whose program is not eligible for the job cannot apply | PASS | 19.2 |
| TC-TRG-05 | BR-11 | A student with more active backlogs than the job allows cannot apply | PASS | 19.1 |
| TC-TRG-06 | BR-11 | A student already placed in the cycle cannot apply to another job in it | PASS | 13.9 |
| TC-TRG-07 | BR-11 | An eligible student's application is accepted with status APPLIED | PASS | 20.8 |
| TC-TRG-08 | Offer rule | An offer cannot be made on a REJECTED or WITHDRAWN application | PASS | 6.8 |
| TC-TRG-09 | Offer rule | Making an offer moves the application to OFFERED (AFTER INSERT trigger) | PASS | 18.5 |
| TC-TRG-10 | BR-12 | Accepting an offer declines other offers, withdraws open applications, frees interview slots and logs it<br>Scenario: PGP62-084 holds a pending offer for job 19, gets a second offer for job 21, and is shortlisted with a scheduled interview for job 22. | PASS | 44.6 |
| TC-TRG-11 | BR-14 | After acceptance the derived status in v_placement_status becomes PLACED | PASS | 40.8 |
| TC-TRG-12 | BR-12 | An offer that was declined (manually or by the cascade) cannot later be accepted | PASS | 35.2 |
| TC-TRG-13 | BR-12 | A student placed in summer can still apply to and accept a final-placement offer | PASS | 36.1 |
| TC-TRG-14 | BR-12 | Accepting an offer after its response deadline is rejected | PASS | 11.6 |
| TC-TRG-15 | Procedure | sp_expire_stale_offers() expires overdue pending offers and logs each one | PASS | 20.4 |
| TC-TRG-16 | Function | fn_apply_for_job reports an unknown roll number clearly | PASS | 7.1 |

## 6.4 Views and queries

The five analytical queries and both views return provably correct answers.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-QRY-01 | Queries | All five queries are present in 05_queries.sql and parse | PASS | 0.2 |

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-QRY-02 | BR-14 | v_placement_status says PLACED exactly when the student has an accepted offer in that cycle | PASS | 51 |
| TC-QRY-03 | Views | v_job_details has one row per job with exactly one of CTC / stipend | PASS | 13.5 |
| TC-QRY-04 | Q1 | Every company returned has >= 20 applicants and an offer rate below the campus average | PASS | 27.4 |
| TC-QRY-05 | Q2 | The talent match for job 25 returns the 8 expected students, led by DBA12-039 | PASS | 23.7 |
| TC-QRY-06 | Q2 | Double NOT EXISTS gives the same answer as division-by-counting for every job<br>Cross-checks the two classic SQL formulations of relational division. | PASS | 55.3 |
| TC-QRY-07 | Q3 | DENSE_RANK puts the program's highest CTC at rank 1 and above_avg_by equals CTC minus the program average | PASS | 31 |
| TC-QRY-08 | Q4 | The running total never decreases and ends at the program's number of final placements | PASS | 15.8 |
| TC-QRY-09 | Q5 | Every 'hidden gem' is unplaced, out-scored the selected students' average CGPA, and failed or missed a round | PASS | 53.1 |

## 6.5 Data invariants

The whole seeded dataset satisfies every rule and is reproducible.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-DAT-01 | Data | The seed produces the documented volumes (reproducible with setseed) | PASS | 18.6 |
| TC-DAT-02 | Data | Rebuilding the database twice produces byte-identical application and offer data | PASS | 1117.1 |
| TC-DAT-03 | BR-12 | No student holds more than one accepted offer in any cycle | PASS | 7.4 |
| TC-DAT-04 | BR-8 | No student has two overlapping interview slots | PASS | 12.3 |
| TC-DAT-05 | BR-11 | Every application meets its job's program, CGPA, backlog and deadline rules | PASS | 12.3 |
| TC-DAT-06 | Data | Status timestamps never precede the event they follow | PASS | 9.6 |
| TC-DAT-07 | Offer rule | Every application that received an offer has status OFFERED | PASS | 8.9 |
| TC-DAT-08 | BR-4 | Every job has exactly one subtype row (disjoint and total) | PASS | 7.8 |
| TC-DAT-09 | Data | A student only sits round k of a job after clearing round k-1 | PASS | 5 |

## 6.6 Transactions & concurrency

Atomicity, isolation under two real concurrent sessions, deferred consistency at COMMIT.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-TXN-01 | Atomicity | Rolling back an acceptance undoes the entire trigger cascade | PASS | 32.9 |
| TC-TXN-02 | Isolation | A second session accepting another offer for the same student waits on the student row lock | PASS | 861.8 |
| TC-TXN-03 | Isolation | When the first session commits, the waiting session fails and the student ends with exactly one accepted offer<br>Session B waits on the lock, then re-reads its offer, finds it DECLINED by A's trigger, and aborts. | PASS | 817.4 |
| TC-TXN-04 | Consistency | COMMIT fails for a job without a subtype and leaves the database unchanged | PASS | 106.4 |

## 6.7 Performance

Index effectiveness measured with EXPLAIN ANALYZE on 600,000 rows.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-PRF-01 | Indexing | A composite B-tree index on (job_id, status) replaces the sequential scan and is at least 10x faster | PASS | 1065 |
| TC-PRF-02 | Indexing | A partial index on only the open rows is far smaller than the full composite index and still used | PASS | 988.2 |
| TC-PRF-03 | Indexing | Without the leading column (job_id) the composite index cannot seek: the status-only query touches 50x more blocks<br>Observed on PostgreSQL 18: the planner may still use the index, but only by reading all of it (or it falls back to a sequential scan), never by seeking. | PASS | 965.9 |

## 6.8 Security

Least-privilege role (DCL) and SQL-injection behaviour, sandboxing included.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-SEC-01 | DCL | placeiq_readonly can SELECT the two tables it was granted | PASS | 6.4 |
| TC-SEC-02 | DCL | placeiq_readonly is denied SELECT on tables it was not granted (offer) | PASS | 5.8 |
| TC-SEC-03 | DCL | placeiq_readonly cannot INSERT, UPDATE, DELETE or DROP | PASS | 9.6 |
| TC-SEC-04 | DCL | placeiq_readonly cannot call pg_read_file (unlike the postgres superuser) | PASS | 1.6 |
| TC-SEC-05 | Injection | The deliberately unsafe search returns all 180 students for x' OR '1'='1 | PASS | 52.2 |
| TC-SEC-06 | Injection | The parameterised search returns 0 rows for the same payload | PASS | 47.2 |
| TC-SEC-07 | Injection | A UNION payload on the unsafe path exposes phone numbers, showing the real risk | PASS | 48.2 |
| TC-SEC-08 | Defence in depth | A DROP TABLE payload is refused by the read-only transaction and the table survives | PASS | 160.6 |
| TC-SEC-09 | Parameterisation | An injection string passed to fn_apply_for_job is treated as a literal roll number | PASS | 8.1 |

## 6.9 REST API

Endpoints return correct data and surface database rule violations clearly.

| ID | Req. | Test case | Result | ms |
|---|---|---|---|---|
| TC-API-01 | API | GET /api/health reports the placeiq database as reachable | PASS | 112.3 |
| TC-API-02 | API | GET /api/summary returns the final-cycle KPIs (30 of 177 placed) | PASS | 90.3 |
| TC-API-03 | API | Student search is case-insensitive and matches names | PASS | 70.1 |
| TC-API-04 | API | An unknown roll number returns HTTP 404 | PASS | 51.1 |
| TC-API-05 | API | Jobs list has 25 jobs; job 19's funnel shows 70 applicants | PASS | 152 |
| TC-API-06 | API | All five analytical queries run through the API in a read-only transaction | PASS | 380.5 |
| TC-API-07 | API | A rejected application returns HTTP 409 with the SQLSTATE and the trigger that raised it | PASS | 61.3 |
| TC-API-08 | API | Accepting an offer via the API returns the cascade the trigger performed | PASS | 243.1 |
| TC-API-09 | API | Declining an offer that is not pending returns HTTP 409 | PASS | 144.5 |
| TC-API-10 | UI | The single-page UI and its assets are served | PASS | 144.1 |
| TC-API-11 | API | POST /api/admin/reset rebuilds the seeded database | PASS | 1228.1 |

# 7. Focus areas

## 7.1 Concurrency (TC-TXN-02, TC-TXN-03)

These tests open two real database connections. A second pending offer is committed for student PGP62-084, then:

1. Session A calls sp_accept_offer for the first offer and does not commit. It holds the student row lock.
2. Session B sets lock_timeout = 700 ms and calls sp_accept_offer for the second offer. It fails with SQLSTATE 55P03 (lock not available): B was blocked by A (TC-TXN-02).
3. In TC-TXN-03, B runs in a thread with no timeout. The test checks that B is still blocked after 0.6 s, then commits A.
4. B wakes, re-reads its offer under the lock, finds it DECLINED by A's cascade, and fails with 'only a PENDING offer can be accepted'.
5. Final check from a third connection: the student has exactly one accepted offer.

## 7.2 Performance (TC-PRF-01 to 03)

**Table 3.** *Measured on a 600,000-row archive, best of three EXPLAIN ANALYZE runs*

| Measurement | Value |
|---|---|
| Sequential scan | 62.654 ms |
| Composite B-tree (job_id, status) | 0.029 ms (2,160× faster) |
| Partial index (open rows only) | 0.021 ms |
| Full composite index size | 4,488 KB |
| Partial index size | 176 KB (26× smaller) |
| Blocks read: seek on job_id | 3 |
| Blocks read: status-only filter | 4,383 |

Assertions are relative (at least 10× faster, at least 10× smaller, at least 50× more blocks) so that the tests are robust to machine speed.

## 7.3 Security (TC-SEC-01 to 09)

The least-privilege role can read only student and person; it is denied other tables, all writes, DROP, and pg_read_file. Through the API, the unsafe search returns all 180 students for x' OR '1'='1 and 180 phone numbers for a UNION payload, while the parameterised search returns 0 rows for the same input. A DROP TABLE payload is refused with SQLSTATE 25006 and the table is verified intact. Passing an injection string to fn_apply_for_job is treated as a literal roll number.

## 7.4 Query correctness

Q2 (relational division) is compared with the division-by-counting formulation for all 25 jobs; the sets are identical. Q1's rows are re-checked against an independently computed campus-wide offer rate; Q3's above_avg_by is recomputed; Q4's running total must be non-decreasing and end at the number of final placements per program; each Q5 row must be unplaced, out-score the selected students' average CGPA and show a failed or missed round.

# 8. Defect and issue log

All issues found during development and testing, including those in the tests themselves.

| ID | Found by | Symptom | Root cause | Resolution | Status |
|---|---|---|---|---|---|
| D-01 | Phase 2 smoke test | Inserting rows with explicit ids left the IDENTITY sequences at 1; the next default insert collided with an existing key. | Explicit ids do not advance a sequence. | setval() for every identity column at the end of 04_data.sql; verified by TC-SCH-07. | Fixed |
| D-02 | Query review (Q4) | Some offers were accepted on 29 Oct, after the dataset's 'as of' date of 28 Oct. | Interview calendar started one day too late. | Final-placement interview days shifted one day earlier. | Fixed |
| D-03 | Live demo dry run | After a pre-placement offer was auto-declined, that job's scheduled interview stayed booked. | Cascade only freed slots of WITHDRAWN applications, not of OFFERED ones whose offer was declined. | Cascade now deletes every SCHEDULED interview for the student's other jobs in the cycle; TC-TRG-10. | Fixed |
| D-04 | Design review of concurrency | Two sessions accepting offers for the same student could deadlock. | Procedure locked the offer row before the student row. | sp_accept_offer locks the student row first (single lock order); TC-TXN-02/03. | Fixed |
| D-05 | UI visual check | Dashboard panels in the same row were vertically misaligned. | A sibling-margin CSS rule applied inside grids. | Spacing applied only between top-level blocks. | Fixed |
| D-06 | UI visual check (mobile) | Page overflowed horizontally at phone width. | CSS grid items default to min-width:auto. | minmax(0, 1fr) and min-width:0 on layout children. | Fixed |
| D-07 | First test run, TC-PRF-03 | PostgreSQL 18 used the (job_id, status) index for a status-only filter, contradicting the demo script. | The textbook rule is about seeking; PG 18 can read the whole index. | Test now compares blocks read (3 vs 4,383); demo wording corrected. | Fixed (documentation) |
| D-08 | First test run, TC-DAT-01 | A row created by one test persisted into later tests. | psycopg's conn.transaction() on an idle connection COMMITs; it is a savepoint only inside an open transaction. | The db fixture opens the outer transaction immediately. | Fixed (test harness) |
| D-09 | First test run, TC-SCH-03/04 | Expected 34 foreign keys (actual 30); composite FKs reported as unindexed although a UNIQUE index on their leading column covers them. | Wrong expectations in the tests. | Expected count corrected; coverage rule accepts a unique leading-column index. | Fixed (test) |

# 9. Running the tests

```
# from the Project folder, with PostgreSQL running and the app venv created
app\.venv\Scripts\pip install pytest httpx
app\.venv\Scripts\python -m pytest tests -v                      # human-readable
app\.venv\Scripts\python -m pytest tests --junitxml=tests\results\junit.xml

# connection settings: PGHOST, PGPORT, PGDATABASE, PGUSER, PGPASSWORD
# psql location (for the rebuild fixture): PLACEIQ_PSQL
```

Outputs: tests/results/results.json (every test with id, requirement, outcome and duration), tests/results/junit.xml (for CI tools) and tests/results/performance.json (timings).

# 10. Conclusion and limitations

All 91 tests pass, covering every business rule, the correctness of all five queries, ACID behaviour under real concurrency, measured index performance and SQL-injection defence. Development testing found nine issues, all resolved and recorded above.

## Not covered

- Automated browser tests of the UI: the UI was verified through the API tests, visual checks at desktop and mobile widths, and the scripted demo recording.
- Load and stress testing beyond the 600,000-row indexing experiment.
- Crash-recovery (durability) testing, which is a guarantee of the PostgreSQL engine rather than of this schema.